

# **Link Control Protocol**

(LCP)

# What is LCP?

LCP stands for Link Control Protocol.

- ▶ It is part of the PPP (Point-to-Point Protocol) used in data communication.
- ▶ Its main job is to establish, configure, test, and terminate a data link between two devices.

# Purpose of LCP

- ▶ To establish a connection between two devices
- ▶ To configure link options like authentication
- ▶ To detect errors and terminate links when necessary
- ▶ To support different network layer protocols over the same link

# Key Functions of LCP

1. Link Establishment – Starts and negotiates parameters for the link
2. Link Configuration – Sets options like:
  - Maximum frame size
  - Authentication method (PAP/CHAP)
  - Compression options
3. Link Maintenance – Monitors link quality, detects errors, and keeps it alive
4. Link Termination – Closes the link when no longer needed

# LCP Packet Structure (Basic Idea)

- Each LCP packet contains:

| Field      | Purpose                                            |
|------------|----------------------------------------------------|
| Code       | Type of message (Configure, Terminate, Echo, etc.) |
| Identifier | Matches requests and replies                       |
| Length     | Size of the packet                                 |
| Data       | Options for configuration or information           |

# Common LCP Messages

- ▶ Configure-Request → Propose options for the link
- ▶ Configure-Ack → Accept proposed options
- ▶ Configure-Nak / Configure-Reject → Reject or modify options
- ▶ Terminate-Request / Terminate-Ack → End the link
- ▶ Echo-Request / Echo-Reply → Check if link is active
- ▶ Code-Reject → Reject unknown LCP codes

# LCP in PPP

- ▶ PPP (Point-to-Point Protocol) is widely used in dial-up connections, DSL, and VPNs
- ▶ LCP is the first step in PPP before sending actual data (like IP packets)
- ▶ After LCP establishes the link, NCP (Network Control Protocol) handles the network layer protocols

# Advantages of LCP

- ▶ Flexible: can work with different network protocols
- ▶ Provides error detection and link management
- ▶ Supports authentication for secure connections